SOFTWARE ARCHITECTURE

Adelpha: local-first MRI engineering platform

Desktop integration of scanner engineering, digital-twin telemetry, MRI acquisition, and optional AI-assisted workflows.

macOS · Linux · Windows

1 DESKTOP BOUNDARY

Tauri v2 / Rust application shell

Native windows · paths · PTY · Python sidecar lifecycle

USER-FACING APPLICATIONS

React / TypeScript frontend

Three.js visualization · IndexedDB persistence

Engineering Studio

CAD assembly, inspection, and simulation part selection.

GLB model workflow

Digital Twin

Scanner dashboard, telemetry, forecasting, and system assessment.

Logging · terminal · camera

Imaging Console

Exam queue, protocol editing, sequence control, and image review.

Study and image viewers

Shared scanner context, settings, and persisted workspace layout

REST · WebSocket · SSE
Local gateway + per-session token

2 LOCAL SERVICE LAYER

Python supervisor / sidecar runtime

Twin and MRI APIs are mounted locally; the optional Agents service is reverse-proxied.

Authenticated loopback gateway
127.0.0.1 · dynamic port

LOCAL TRUST BOUNDARY

DTAM Twin API

`/api/dtam`

- Thermal → B_0 / f_0 estimates
- EMI and RF state
- Forecast and assessment
- State provenance labels

Read-only telemetry interface

MRI FastAPI façade

`/api/mri`

- MRI4ALL folder queue
- scan.json / PREPARED
- SequenceBase · PyPulseq
- Acquisition + reconstruction

Python: FLOCRA / MaRCoS

Google ADK Agents

OPTIONAL

`/api/agents`

- Sessions and chat
- Streaming and artifacts
- Lazy child process
- User-provided API key

Reverse-proxied by supervisor

Sensor / adapter state
Measured or simulated

Python hardware I/O
Sequence output / raw data

Model requests / responses
Optional external API

Scanner adapters

Sensors or simulated plant

Thermal · EMI · RF channels

MRI scanner hardware

Red Pitaya: MaRCoS server

FPGA · RF and gradients

Gemini model API

Optional model provider

Configured in AI & Agents

3 PHYSICAL AND EXTERNAL INTERFACES

Scanner hardware, sensors, and model provider

API / data exchange — Optional component or external service

API, application programming interface; EMI, electromagnetic interference; RF, radio frequency.

Figure. Adelpha software architecture. The Tauri desktop shell hosts three React workspaces and manages an authenticated Python sidecar. Local APIs connect digital-twin services, MRI acquisition hardware, and optional agent-based model services.